

Max Teeple

Scottsdale, Arizona · max.teeple@gmail.com · maxteeple.com · github.com/Max-Teeple

I work on perception and autonomy for agricultural machines: how they see, where they are, and how they steer. Lately that has been lidar benchmarking, wide-baseline stereo and the algorithms we write for it, and filtering and detection on 3D sensor data.

PROFESSIONAL EXPERIENCE

Research Engineer II

May 2023 – Present

CNH Industrial · Scottsdale, AZ

- Own sensor selection for perception, benchmarking lidar systems and OEMs for autonomous and semi-autonomous agricultural vehicles.
- Specify wide-baseline stereo and develop the in-house stereo algorithms that run on the vehicle.
- Take 3D sensor data through filtering and detection so the autonomy stack has structure and tracks it can use.
- Design, simulate, and tune model-based vehicle controllers, and carry that software onto the machine.
- Integrate the perception and control stack into production platforms with the teams around the vehicle.

Research Engineer

May 2022 – May 2023

Raven Industries · Scottsdale, AZ

- Developed a loosely coupled inertial navigation system from chip selection through product launch.
- Built the sensor-fusion path the navigation estimate depends on, and validated the embedded software in real time on precision-agriculture machines.
- Modeled vehicle guidance and stability around that navigation solution.

Graduate Research Assistant

Fall 2020 – Spring 2022

Iowa State University · Ames, IA

- Developed model-based control for dynamic vehicle suspensions in partnership with John Deere.
- Carried the work through control design, simulation, and hardware-in-the-loop validation.
- Wrote the technical documentation and contributed to peer-reviewed research.

Product Engineering Intern

Summer 2021

John Deere · Waterloo, IA

- Built predictive machine-learning models to analyze engine emission signals.

Product Engineering Intern

Summer 2020

John Deere · Des Moines, IA

- Developed MATLAB and Simulink models for prototype machine-health systems.

Product Engineering Intern

Summer 2019

John Deere · Des Moines, IA

- Modeled electromechanical nozzle control systems in Simulink.

EDUCATION

M.S., Mechanical Engineering

Fall 2020 – Spring 2022

Iowa State University · Ames, IA

B.A., Physics and Computer Science

Fall 2016 – Spring 2020

Coe College · Cedar Rapids, IA

Minor in Mathematics

TECHNICAL SKILLS

Programming Python, Java, MATLAB, Simulink, C, C++

Tools Creo, SOLIDWORKS

Focus Perception, autonomy, lidar and stereo, 3D filtering and detection, sensor fusion, inertial navigation, and vehicle controls